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Data Science

AgroAssist – Farm Advisor

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Project Guide
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1. SDG

Sdg 15: Life on Land - Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss

2. Introduction

Problem Identified:

• Small & marginal farmers lack access to structured crop management tools
• Information on cultivation, scheduling, and weather is scattered
• No single platform to manage farmer profiles, tasks, and crop records
• Poor productivity, resource wastage, and no proactive weather response

Solution Proposed:

• AgroAssist: A full-stack web & mobile agriculture management platform
• Django REST API backend + Flutter frontend for cross-platform access
• Role-based access for Admins and Farmers with secure token-based authentication
• Unified modules: Crops, Farmers, Tasks & Weather in one platform

3. Objectives

1)To provide time-based crop task reminders that help farmers perform farming activities at the right time.
2)To alert farmers about common crop-related mistakes and reduce losses caused by incorrect farming practices.
3)To calculate required farming inputs such as seeds, fertilizers, and water based on land area.
4)To help farmers compare crops easily and select the most suitable crop for their resources and season.
5)To explain crop growth stages visually for better understanding with minimal text.

4)scope

• Full-stack monorepo system: Django REST backend + Flutter frontend application.
• Supports four primary domains: Crops, Farmers, Tasks, and Weather.
• Accessible as a web application and Android mobile app.
• Role-based access control differentiates Admin (full access) and Farmer (restricted) roles.
• Can be implemented in agricultural communities, NGOs, and agri-tech platforms across India.
• Applicable to agricultural extension workers promoting modern farming techniques.
• Extendable to government agricultural programs and crop insurance schemes.
• Current scope does not include real-time satellite imagery, IoT sensors, or e-commerce.

5. Feature /Functionality

• **Feature 1**: Full CRUD operations on crop records with filtering by soil type & season, plus crop recommendations based on farmer inputs
• **Feature 2:** Registration and profile updates for farmer records, with crop linkage to track cultivation progress per farmer.
• **Feature 3:** Admin creates tasks and assigns them to farmers. Lifecycle tracking: Pending → In Progress → Completed with backend validation.
• **Feature 4:** Backend stores weather advisories from external API. Location-based alerts and forecasts delivered to the Flutter frontend for timely farmer guidance.

6. Outcome of Project

• Fully functional full-stack web & mobile application (browser + Android)
• Flutter frontend with role-aware navigation, real-time API integration & session management
• Weather alert module displaying location-based advisories to farmers
•CSV bulk import facility for populating crop and farmer data efficiently
• Working task lifecycle management system enforcing status transitions and validation
• Well-structured Django REST API with modular apps: Crops, Farmers, Tasks, Weather, Auth

7. Technology Stack

Frontend:
- Flutter (Dart 3.x)

Backend:
- Django + DRF

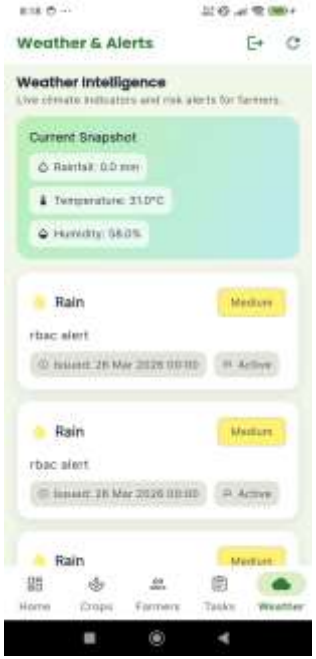
Database:
- SQLite (default), PostgreSQL(production)

Authentication:
- Token-Based Auth

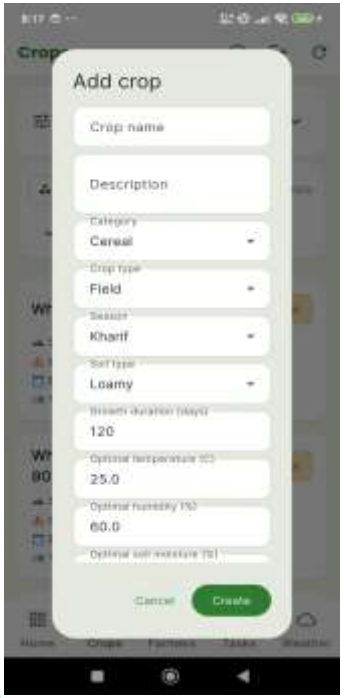
Deployment:
- Vercel + Local Django

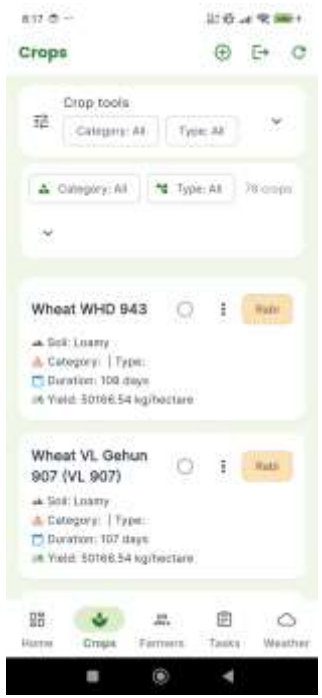
UI Mockups

AgroAssist Dash...
English
8
Pending Tasks
0
Active Alerts
Quick Actions
Jump directly into your most used workflows
Browse Crops
View all available crops and guides
Manage Farmers
View and manage farmer profiles
View Tasks
Check and manage farming tasks
Weather Alerts
View weather forecasts and alerts
Home
Crops
Farmers
Tasks
Weather


Weather & Alerts
Weather Intelligence
Live climate indicators and risk alerts for farmers
Current Snapshot
Rainfall: 0.0 mm
Temperature: 31.0°C
Humidity: 58.0%
Rain
Medium
rbac alert
Issued: 28 Mar 2026 00:00
Active
Rain
Medium
rbac alert
Issued: 28 Mar 2026 00:00
Active
Rain
Medium
Home
Crops
Farmers
Tasks
Weather

Tasks
Task Operations
Track work progress and close tasks quickly
All
Pending
In Progress
Completed
Spray micronutrients
Completed
Farmer: Imran Khan
Crop: Tomato
Due: 20 Feb 2026 (00)
Priority: 6
Medium
Apply foliar micronutrient dose in early morning.
First irrigation
In Progress
Farmer: Sunita Sharma
Crop: Wheat
Due: 27 Feb 2026 (-55d)
Create Task
Priority: 8
High
Home
Crops
Farmers
Tasks
Weather


Add crop
Crop name
Description
Category
Cereal
Crop type
Field
Season
Kharif
Soil type
Loamy
Growth duration (days)
120
Optimal temperature (°C)
25.0
Optimal humidity (%)
60.0
Optimal soil moisture (%)
Cancel
Create


Crops
Crop tools
Category: All
Type: All
Category: All
Type: All
78 crops
Wheat WHD 943
Rabi
Soil: Loamy
Category: | Type:
Duration: 108 days
Yield: 50166.54 kg/hectare
Wheat VL Gehun 907 (VL 907)
Rabi
Soil: Loamy
Category: | Type:
Duration: 107 days
Yield: 50166.54 kg/hectare
Home
Crops
Farmers
Tasks
Weather

Wheat VL Gehun 907 (VL 907) - Crop Guide
Season: Rabi
Soil: Loamy
Temperature: 20.5-3C
Watering: -
Fertilizer: Phosphorus,Nitrogen
Disease: -
Harvest: -
Input estimate by land area
Area: 1.00 ha
Seeds: 55.0 kg
Fertilizer: 220.0 kg
Water / week: 250000 liters
Growth stages
Sowing
Vegetative
Flowering
Harvest
Stage-wise care tasks
Close

8. Block Diagram

9)Conclusion

- **Successful Deployment:** Developed **AgroAssist**, a full-stack platform effectively solving real-world agricultural management challenges.
- **Robust Architecture:** Leveraged **Django and Flutter** to build a scalable, secure system with role-based access control.
- **Technical Growth:** Mastered **REST API design** and agile development through end-to-end project execution.
- **Future Vision:** Poised for expansion into **IoT sensors, AI yield prediction,** and cloud-based scaling.

Thank You...!!